SECTION 2

What Can Be Taken?

Three cases, from 1876 to today

Most theft is easy to recognise. Something leaves its owner without permission, and the owner no longer has it. The three cases here are harder. In each, something small moved: seeds from a forest, blood from a village, photos from the internet. Yet it is hard to say what the owner lost. Brazil kept its rubber trees. The donors gave their blood willingly. The original photos stayed online.

What changed was what each thing became: plantations on another continent, data for studies the donors say they never agreed to, a search engine for faces. Those who call these cases theft point to what the thing became, not only to what moved.

The cases span 150 years, from the age of empire to the age of the internet. Each panel sets out what happened, who calls it theft and why, and how the other side answers.

Audio guide · Section 2
Scan to listen

SECTION 2 · STORY 1

SEVENTY THOUSAND SEEDS

Henry Wickham · the Amazon to Kew Gardens, London · 1876

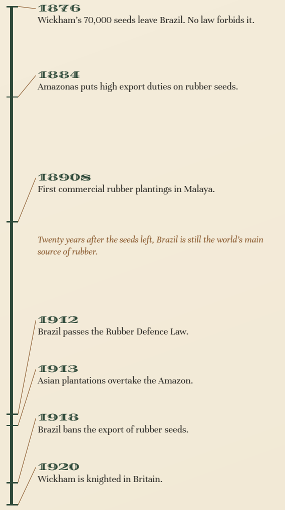
In 1876, Henry Wickham, an Englishman living in the Amazon, shipped about 70,000 rubber seeds to Kew Gardens in London, paid by Britain's India Office. By his own later account, he told customs they were "exceedingly delicate botanical specimens". Brazil was then the world's main source of rubber. The seedlings went on to British Asia. For twenty years little changed; then plantation labour, investment and the demand for bicycle and car tyres turned them into an industry. By 1913 Asian plantations had overtaken the Amazon, and Brazil's rubber boom ended. Kerala's rubber trees descend largely from the same seeds.

In Brazil, Wickham is remembered as "the father of bio-piracy": someone who profited from Brazil's plants without its consent. Wickham called it patriotism. No Brazilian law in 1876 banned taking the seeds, Brazil kept its trees, and the plantations took decades of work to succeed. Britain knighted him in 1920.

Brazil kept its rubber trees.
What, if anything, did Britain take?

IT TOOK DECADES

1876–1920, drawn to scale



1876 Wickham's 70,000 seeds leave Brazil. No law forbids it.

1884 Amazonas puts high export duties on rubber seeds.

1890s First commercial rubber plantings in Malaya.

Twenty years after the seeds left, Brazil is still the world's main source of rubber.

1912 Brazil passes the Rubber Defence Law.

1913 Asian plantations overtake the Amazon.

1918 Brazil bans the export of rubber seeds.

1920 Wickham is knighted in Britain.

Timeline · Made for this exhibition

SECTION 2 · STORY 2

WHAT THE BLOOD WAS FOR

The Havasupai Tribe and Arizona State University · Grand Canyon, USA · 1990–2010

The Havasupai are a Native American tribe of about 650 people, living deep in the Grand Canyon. From the 1960s, type 2 diabetes spread through the tribe at an extraordinary rate. It led to amputations, even among the young, and sent many out of the canyon for dialysis. In late 1989 a tribal elder, Rex Tilousi, asked an anthropologist at Arizona State University (ASU) whom the tribe trusted whether he knew a doctor who could help. From 1990, ASU researchers collected blood from more than 200 members. Tribal leaders understood the study was about diabetes; the written consent form spoke more broadly, of "behavioral/medical disorders".

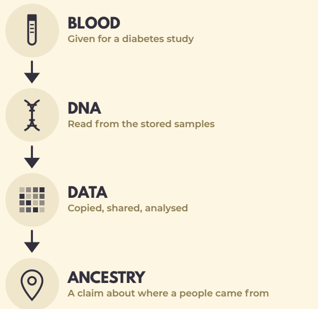
The samples went on to support 23 papers and dissertations. Fifteen had nothing to do with diabetes; they studied schizophrenia, inbreeding and the tribe's origins. The researchers saw these as part of the same search for what causes disease: the lead researcher said donors were told the work looked at illnesses "such as diabetes, schizophrenia, depression", and inbreeding can raise the risk of illness. The tribe saw it differently. One paper suggested their ancestors came from Asia; their own history says they began in the canyon. Tribal council member Carletta Tilousi said: "people got degrees and grants, and they never asked our permission."

The lead researcher, Therese Markow, defended the work as ethical: genetic research, she argued, often advances through studies that do not seem tied to any one disease. In 2010, ASU's governing board settled. It paid $700,000 to 41 tribe members, returned the samples and apologised. No court ever ruled on whether the research broke the law. The blood went home. The published papers did not.

They gave blood for a health study.
How much permission travelled with it?

DIAGRAM

FROM BLOOD TO ANCESTRY



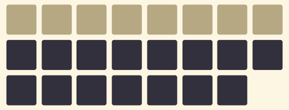
BLOOD
Given for a diabetes study

DNA
Read from the stored samples

DATA
Copied, shared, analysed

ANCESTRY
A claim about where a people came from

23 PAPERS & DISSERTATIONS



- ▨ 8 on diabetes
- ■ 15 on schizophrenia, inbreeding or the tribe's origins

From blood to ancestry (diagram)
Made for this exhibition
Blood becomes DNA, DNA becomes data, and data can answer questions no one asked when the blood was given.

Section 2 · Story 3

A Search Engine for Faces

Clearview AI · New York, USA · 2017 to today

Clearview AI, an American company founded in 2017, copies publicly posted photos from social media, news sites and other websites, without telling the people in them. Its software turns each face into a faceprint, a list of numbers describing its geometry. One uploaded photo can then find other photos of the same face online. By its own count, it holds more than 100 billion images, and it sells searches mainly to police and government agencies.

Regulators in Canada, Australia and several European countries found it broke their privacy laws. Canada's privacy commissioner said: "What Clearview does is mass surveillance and it is illegal." Illinois law draws its line at the faceprint: it protects scans of face geometry, but not photographs.

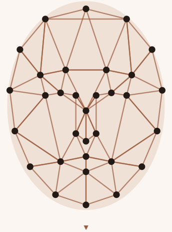
Clearview's co-founder, Hoan Ton-That, said: "All the data inside Clearview AI is publicly available, as it is with Google." The company says police use it to identify suspects and abused children. It admits no wrongdoing, and has not paid several European fines.

Status as of October 2026.

Nobody lost a photograph.
So what, if anything, was taken when the people in them became searchable?

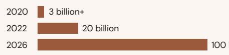
Faceprint

A face as numbers



001	+0.1450
002	-0.0337
003	-0.1886
004	-0.0439
005	-0.0716
006	-0.1260
007	-0.0025
008	+0.1255
009	-0.1370
010	+0.0295
011	+0.0688
⋮ 512	+0.0731

Images in the database

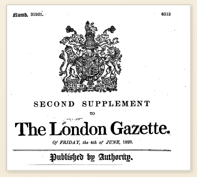


2020	3 billion+
2022	20 billion
2026	100 billion+

Images, not people. The 2026 figure is Clearview's own claim.

A face as numbers (faceprint)
Made for this exhibition · Values are illustrative
Canada's privacy commissioners described each Clearview faceprint as a 'vector' of 512 data points.

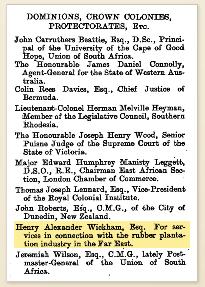
BRITAIN'S VERDICT **1920**



Numb. 31931. 6313

SECOND SUPPLEMENT
TO
The London Gazette.
Of FRIDAY, the 4th of JUNE, 1920.
Published by Authority.

Second Supplement to The London Gazette, issue 31931, page 6313 · Birthday Honours: Knights Bachelor

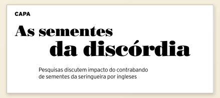


DOMINIONS, CROWN COLONIES, PROTECTORATES, Etc.

John Carruthers Beattie, Esq., D.Sc., Principal of the University of the Cape of Good Hope, Union of South Africa.
The Honourable James Daniel Connolly, Agent-General for the State of Western Australia.
Colin Rees Davies, Esq., Chief Justice of Bermuda.
Lieutenant-Colonel Herman Melville Heyman, Member of the Legislative Council, Southern Rhodesia.
The Honourable Joseph Henry Wood, Senior Puisne Judge of the Supreme Court of the State of Victoria.
Major Edward Humphrey Manisty Leggett, D.S.O., R.E., Chairman East African Section, London Chamber of Commerce.
Thomas Joseph Lennard, Esq., Vice-President of the Royal Colonial Institute.
John Roberts, Esq., C.M.G., of the City of Dunedin, New Zealand.
Henry Alexander Wickham, Esq. For services in connection with the rubber plantation industry in the Far East.
Jeremiah Wilson, Esq., C.M.G., lately Post-master-General of the Union of South Africa.

"For services in connection with the rubber plantation industry in the Far East"
1920 · Knighthood, The London Gazette, 5 June 1920, p. 6315 · Reproduction from thegazette.co.uk
Britain's verdict: the official notice knighting Henry Alexander Wickham. It names the industry his seeds began. It does not mention Brazil.

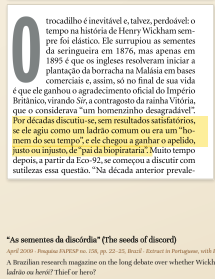
BRAZIL'S VERDICT **2009**



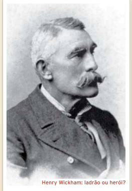
CAPA

As sementes da discórdia

Pesquisas discutem impacto do contrabando de sementes da seringueira por ingleses



O trocadilho é inevitável e, talvez, perdoável: o tempo na história de Henry Wickham sempre foi elástico. Ele surrupiou as sementes da seringueira em 1876, mas apenas em 1895 é que os ingleses resolveram iniciar a plantação da borracha na Malásia em bases comerciais e, assim, só no final de sua vida é que ele ganhou o agradecimento oficial do Império Britânico, virando *Sir*, a contragosto da rainha Vitória, que o considerava "um homenzinho desagradável". Por décadas discutiu-se, sem resultados satisfatórios, se ele agiu como um ladrão comum ou era um "homem do seu tempo", e ele chegou a ganhar o apelido, justo ou injusto, de "pai da biopirataria". Muito tempo depois, a partir da Eco-92, se começou a discutir com sutilezas essa questão. "Na década anterior prevale-



Henry Wickham: ladrão ou herói?

IN ENGLISH

> "For decades there were discussions, with no satisfactory conclusion being reached, as to whether he behaved like a common thief or whether he was a 'man of his time'; either way, he earned the moniker 'the father of bio-piracy'."

Translation from the magazine's own English edition. The Portuguese says justo ou injusto: 'fair or unfair'.

"As sementes da discórdia" (The seeds of discord)
April 2009 · Pesquisa FAPESP no. 158, pp. 22–25, Brazil · Extract in Portuguese, with English
A Brazilian research magazine on the long debate over whether Wickham was a thief or a man of his time. Its photo caption asks: *ladrão ou herói?* Thief or hero?

Pages: Carlos Haag, Pesquisa FAPESP (revistapesquisa.fapesp.br), CC BY-NC-ND. The full article is on the desk.

WHAT DID "YES" MEAN?

THREE VERSIONS OF ONE PERMISSION · 1990–1994

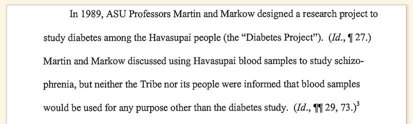
1. WHAT TRIBAL LEADERS WERE TOLD
Diabetes research
The project was described to the Tribal Council again and again as a study of diabetes.

2. WHAT THE WRITTEN FORM SAID
"…to study the causes of behavioral/medical disorders."
The consent form signed by some of the early donors.

3. WHAT LATER DONORS HEARD
Diabetes research
From 1991 to 1994 some blood was taken with spoken, not written, consent, explained as diabetes research.

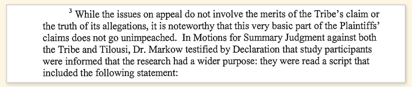
✂ IN COURT, BOTH SIDES QUOTED THE SAME YEARS

THE TRIBE'S CLAIM, AS SUMMARISED IN THE BRIEF

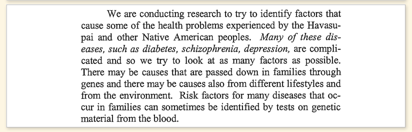


In 1989, ASU Professors Martin and Markow designed a research project to study diabetes among the Havasupai people (the "Diabetes Project"). (*Id.*, ¶ 27.) Martin and Markow discussed using Havasupai blood samples to study schizophrenia, but neither the Tribe nor its people were informed that blood samples would be used for any purpose other than the diabetes study. (*Id.*, ¶¶ 29, 73.)[3]

THE RESEARCHER'S ANSWER: THE SCRIPT DONORS WERE READ



[3] While the issues on appeal do not involve the merits of the Tribe's claim or the truth of its allegations, it is noteworthy that this very basic part of the Plaintiffs' claims does not go unimpeached. In Motions for Summary Judgment against both the Tribe and Tilousi, Dr. Markow testified by Declaration that study participants were informed that the research had a wider purpose: they were read a script that included the following statement:

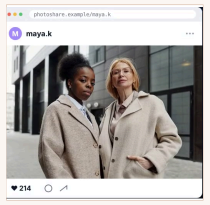


We are conducting research to try to identify factors that cause some of the health problems experienced by the Havasupai and other Native American peoples. *Many of these diseases, such as diabetes, schizophrenia, depression*, are complicated and so we try to look at as many factors as possible. There may be causes that are passed down in families through genes and there may be causes also from different lifestyles and from the environment. Risk factors for many diseases that occur in families can sometimes be identified by tests on genetic material from the blood.

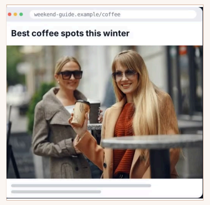
What did "yes" mean? (consent panel)
1990–2007 · Cards made for this exhibition · Court record: Joint Answering Brief, Havasupai Tribe v. Arizona Board of Regents, Arizona Court of Appeals, No. 1 CA-CV 07-0454, pp. 3–4 · Reproduction
Three versions of one permission. Below them, the two sides in court: the Tribe said it was told only about diabetes; Dr Markow said donors were read a script naming 'diabetes, schizophrenia, depression'. No court ruled on which was true.

One face, four places

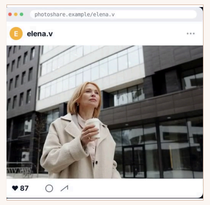
Staged photographs · 2026 · Made for this exhibition with a consenting volunteer



PUBLIC POST
A. A friend's post: she is beside the friend



PUBLIC POST
B. "Best coffee spots this winter"



PUBLIC POST
C. Her own post

NOT POSTED ANYWHERE
D. The photo that gets searched

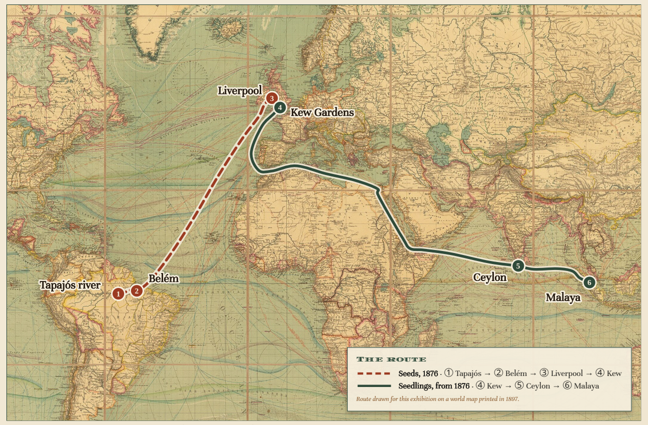
One volunteer in four settings, each photo shared with a different audience. The same stills appear in the face-search video on the desk.

01 **POSTED** Visible online → 02 **COLLECTED** Copied by Clearview → 03 **FACE DATA** Turned into a faceprint → 04 **SEARCHABLE** Found from a new photo

Posted → collected → face data → searchable (diagram) · Made for this exhibition · The four steps that turn someone visible online into someone searchable.

FROM THE TAPAJÓS TO ASIA

1876 to the 1890s



THE ROUTE
- - - - **Seeds, 1876** · ① Tapajós → ② Belém → ③ Liverpool → ④ Kew
—— **Seedlings, from 1876** · ④ Kew → ⑤ Ceylon → ⑥ Malaya

Route drawn for this exhibition on a world map printed in 1897.

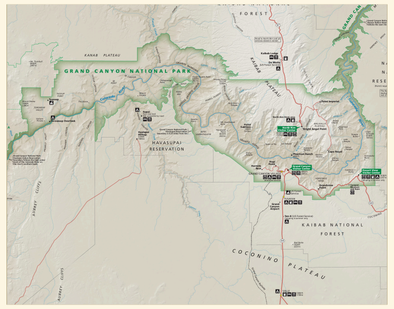
Map: Hermann Berghaus, *Chart of the World on Mercator's Projection* (Gotha: Justus Perthes, 1897), detail. Public domain, via Wikimedia Commons.

From the Tapajós to Asia (map)
1876 to the 1890s · Map printed 1897
The seeds went from the Tapajós river to Kew Gardens in London; the seedlings went on to Ceylon and Malaya. The route is drawn over a world map printed in 1897.

Listen & see the sources
Audio guide · Story 1

MAP

SUPAI, ARIZONA



Supai, Arizona (map)
Grand Canyon National Park map, US National Park Service · Detail
The Havasupai Reservation lies on the canyon's south side, west of Grand Canyon Village. Supai, its village, is reachable only on foot, by mule or by helicopter.

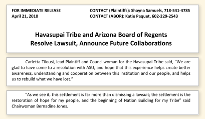
Map: National Park Service, Harpers Ferry Center (public domain), via Wikimedia Commons.

WHAT CAME BACK IN 2010

☑ 151 remaining blood samples
☑ Lab notebooks and family-tree records
☑ $700,000 to 41 tribe members
☑ A formal apology
☑ No new ASU research with the samples

NOT RETURNED
☐ 23 papers and dissertations, still published
☐ Copies and data already passed to other researchers

The blood went home.
The published papers did not.



FOR IMMEDIATE RELEASE
April 21, 2010
CONTACT (Plaintiffs): Shayna Samuels, 718-541-4785
CONTACT (ABOR): Katie Paquet, 602-229-2543

Havasupai Tribe and Arizona Board of Regents Resolve Lawsuit, Announce Future Collaborations

Carletta Tilousi, lead Plaintiff and Councilwoman for the Havasupai Tribe said, "We are glad to have come to a resolution with ASU, and hope that this experience helps create better awareness, understanding and cooperation between this institution and our people, and helps us to rebuild what we have lost."

"As we see it, this settlement is far more than dismissing a lawsuit; the settlement is the restoration of hope for my people, and the beginning of Nation Building for my Tribe" said Chairwoman Bernadine Jones.

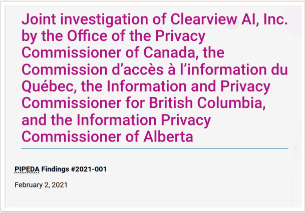
Joint press release, Havasupai Tribe and Arizona Board of Regents, 21 April 2010 · Reproduction

What came back (checklist) · 2010 settlement · Made for this exhibition; quotations from the 2010 press release.

Listen & see the sources
Audio guide · Story 2 · What the Blood Was For

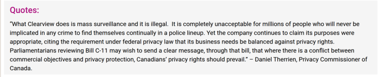
Document

Canada's privacy commissioners



Joint investigation of Clearview AI, Inc. by the Office of the Privacy Commissioner of Canada, the Commission d'accès à l'information du Québec, the Information and Privacy Commissioner for British Columbia, and the Information Privacy Commissioner of Alberta

PIPEDA Findings #2021-001
February 2, 2021



Quotes:
"What Clearview does is mass surveillance and it is illegal. It is completely unacceptable for millions of people who will never be implicated in any crime to find themselves continually in a police lineup. Yet the company continues to claim its purposes were appropriate, citing the requirement under federal privacy law that its business needs be balanced against privacy rights. Parliamentarians reviewing Bill C-11 may wish to send a clear message, through that bill, that where there is a conflict between commercial objectives and privacy protection, Canadians' privacy rights should prevail." – Daniel Therrien, Privacy Commissioner of Canada.

Canada's privacy commissioners
2–3 February 2021 · PIPEDA Findings #2021-001 and news release · Office of the Privacy Commissioner of Canada · Extracts printed from priv.gc.ca
The joint investigation by Canada's federal commissioner and three provincial commissioners found that Clearview's scraping was mass surveillance and broke Canadian law. Clearview disputed the findings and withdrew from Canada.

Source: priv.gc.ca. The full news release is on the desk.

Rulings so far

Year	Where	Decision	Status
2021	Canada	Mass surveillance; illegal	Clearview disputed it
2021	Australia	Stop collecting; destroy Australians' data	Order issued
2022	USA	Most private companies barred	Settlement, no fault admitted
2024	Netherlands	€30.5 million fine	Fine issued
2025	EU	AI Act bans scraping faces to build such databases	In force
2026	Clearview	"More than 100 billion images"	Still operating

Clearview admits no wrongdoing and has not paid several European fines. Status as of October 2026.

Rulings so far (table) · 2021–2026 · Made for this exhibition. Dutch decision: Autoriteit Persoonsgegevens, 3 Sept. 2024 (on the desk).

Listen & see the sources
Audio guide · Story 3 · A Search Engine for Faces

Museum of Theft · Section 2 · What Can Be Taken? · full board

Wall 120 × 90 cm · printed on A3 at 1 : 3.4